

SOLUTION MCQ

1. Solutions defined homogeneous mixture

- a) heterogeneous substances
- b) two or more pure substances
- c) single pure substance
- d) physical separation only

Correct answer: b

2. Solution components solute solvent

- a) equal amounts
- b) solute dissolves solvent
- c) solvent dissolves solute
- d) both same

Correct answer: c

3. Homogeneous mixture uniform composition

- a) throughout
- b) surface only
- c) bottom only
- d) layers

Correct answer: a

4. Heterogeneous mixture non-uniform composition

- a) stable particles
- b) visible separation
- c) transparent
- d) filterable

Correct answer: b

5. Solution particles size less than

- a) 10 nm
- b) 100 nm
- c) 1 nm
- d) 1000 nm

Correct answer: c

6. Solvent present greater

- a) molecular weight
- b) concentration
- c) density
- d) viscosity

Correct answer: b

7. Universal solvent

- a) ethanol
- b) benzene
- c) water
- d) acetone

Correct answer: c

8. Binary solution contains components

- a) one
- b) two
- c) three
- d) many

Correct answer: b

9. Ternary solution components

- a) one solute one solvent

b) two solutes one solvent

c) three solvents

d) four components

Correct answer: b

10. Gas in gas solution example

a) salt water

b) air

c) brass

d) soda

Correct answer: b

11. Solid in liquid solution

a) air

b) sugar water

c) alloys

d) fog

Correct answer: b

12. Liquid in liquid solution

a) ethanol water

b) CO₂ water

c) mercury sodium

d) camphor air

Correct answer: a

13. Aqueous solution solvent

a) benzene

b) water

c) alcohol

d) ether

Correct answer: b

14. Non-aqueous solution solvent

a) water

b) carbon disulphide

c) salt

d) sugar

Correct answer: b

15. Saturated solution no more solute dissolves

a) cooling

b) given temperature

c) heating

d) pressure change

Correct answer: b

16. Unsaturated solution contains solute

a) maximum

b) less than saturated

c) more than saturated

d) equal saturated

Correct answer: b

17. Supersaturated solution unstable

a) precipitates easily

b) very stable

c) transparent

d) coloured

Correct answer: a

18. Concentrated solution large quantity

- a) solvent
- b) solute
- c) water
- d) gas

Correct answer: b

19. Dilute solution small quantity

- a) solvent
- b) solute large solvent
- c) both equal
- d) saturated

Correct answer: b

20. Solubility maximum solute dissolves

- a) minimum solvent
- b) given temperature pressure
- c) room temperature only
- d) boiling point

Correct answer: b

21. Solubility NaCl 25°C equals

- a) 10 g/100g water
- b) 36 g/100g water
- c) 50 g/100g water
- d) 100 g/100g water

Correct answer: b

22. Like dissolves like principle

- a) polar polar

b) nonpolar nonpolar

c) ionic covalent

d) gas liquid

Correct answer: a

23. Solid solubility liquid generally increases

a) cooling

b) temperature rise

c) pressure decrease

d) stirring

Correct answer: b

24. Gas solubility liquid decreases

a) pressure increase

b) temperature increase

c) cooling

d) stirring

Correct answer: b

25. Henry's law gas solubility proportional

a) temperature

b) pressure

c) volume

d) density

Correct answer: b

26. Mass percentage solute mass

a) solvent mass

b) solution mass

c) volume ratio

d) mole ratio

Correct answer: b

27. Volume percentage solute volume

a) solvent volume

b) solution volume

c) mass ratio

d) density

Correct answer: b

28. Hydrated salt contains water

a) loosely bound

b) crystallisation

c) surface adsorbed

d) chemically combined

Correct answer: b

29. Blue vitriol formula

a) NaCl

b) $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$

c) $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$

d) $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$

Correct answer: b

30. Water crystallisation Epsom salt

a) 5

b) 7

c) 2

d) 10

Correct answer: b

31. Hygroscopic substances absorb moisture

- a) dissolve completely
- b) remain solid
- c) form solution
- d) evaporate

Correct answer: b

32. Deliquescent substances absorb moisture

- a) remain crystalline
- b) form saturated solution
- c) dry air
- d) lose water

Correct answer: b

33. Caustic soda shows

- a) hygroscopy
- b) deliquescence
- c) efflorescence
- d) sublimation

Correct answer: b

34. Mixture physical separation

- a) chemical reaction
- b) distillation
- c) chromatography
- d) filtration

Correct answer: b

35. Solution particles pass

- a) filter paper

b) cannot settle

c) both

d) neither

Correct answer: c

36.Solute concentration compared solvent

a) higher

b) lower

c) equal

d) variable

Correct answer: b

37.Dissolution uniform distribution

a) solute solvent

b) solvent solute

c) particles filter

d) gas liquid

Correct answer: a

38.Gas in solid solution

a) air

b) hydrogen palladium

c) soda

d) fog

Correct answer: b

39.Solid in solid solution

a) alloys

b) saltwater

c) air

d) vinegar

Correct answer: a

40. Sea water example

a) non-aqueous

b) aqueous

c) gaseous

d) solid

Correct answer: b

41. 36g NaCl 100g water 25°C

a) unsaturated

b) saturated

c) supersaturated

d) dilute

Correct answer: b

42. Mango juice typically

a) dilute

b) concentrated

c) saturated

d) supersaturated

Correct answer: b

43. Solubility expressed g solute per

a) 100 ml solution

b) 100 g solvent

c) 1 litre solution

d) 1000 g water

Correct answer: b

44. Polar solute dissolves

- a) nonpolar solvent
- b) polar solvent
- c) both
- d) neither

Correct answer: b

45. Nonpolar solute dissolves

- a) water
- b) benzene
- c) alcohol
- d) acetone

Correct answer: b

46. Sugar solubility hot water

- a) decreases
- b) increases
- c) same
- d) zero

Correct answer: b

47. Soda bubbles heating because gas solubility

- a) increases
- b) decreases
- c) same
- d) doubles

Correct answer: b

48. Henry's law constant temperature

- a) solubility inversely proportional pressure

- b) directly proportional pressure
- c) independent pressure
- d) exponential

Correct answer: b

49.5% sugar solution mass contains

- a) 5g sugar 95g water
- b) 5g sugar 100g water
- c) 95g sugar 5g water
- d) 50g sugar 50g water

Correct answer: a

50.10% volume ethanol means

- a) 10ml ethanol 90ml water
- b) 10ml ethanol 100ml water
- c) 10g ethanol 90g water
- d) 100ml ethanol 100ml water

Correct answer: a

51.Copper sulphate pentahydrate loses water heating becomes

- a) blue solution
- b) anhydrous white
- c) deliquescent
- d) hygroscopic

Correct answer: b

52.Gypsum formula

- a) $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$
- b) $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$
- c) $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$

d) $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$

Correct answer: b

53. Concentrated sulphuric acid

a) hygroscopic

b) deliquescent

c) efflorescent

d) anhydrous

Correct answer: a

54. Silica gel used drying agent because

a) deliquescent

b) hygroscopic

c) soluble water

d) reactive

Correct answer: b

55. Solution stable solute particles

a) settle bottom

b) float top

c) remain suspended

d) do not settle

Correct answer: d

56. Fruit salad example

a) homogeneous

b) heterogeneous

c) solution

d) suspension

Correct answer: b

57. Cough syrup typically

- a) heterogeneous
- b) homogeneous
- c) suspension
- d) colloid

Correct answer: b

58. Supersaturated solution disturbed

- a) becomes stable
- b) precipitates crystals
- c) absorbs more solute
- d) evaporates

Correct answer: b

59. Solubility CaCO_3 water very

- a) high
- b) low
- c) moderate
- d) infinite

Correct answer: b

60. Ammonia solubility water

- a) 48 g/100g
- b) 36 g/100g
- c) 91 g/100g
- d) 0.0013 g/100g

Correct answer: a

61. Mass percentage temperature

- a) dependent

b) independent

c) increases

d) decreases

Correct answer: b

62. Volume percentage liquids

a) temperature independent

b) decreases temperature rise

c) increases heating

d) constant

Correct answer: b

63. Hydration ions attract water molecules

a) physically

b) chemically attached

c) loosely bound

d) evaporated

Correct answer: b

64. Anhydrous salt heating hydrated salt

a) gains water

b) loses water

c) changes colour only

d) dissolves

Correct answer: b

65. Green vitriol

a) $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$

b) $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$

c) $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$

d) $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$

Correct answer: b

66. Deliquescence maximum conditions

a) high temperature dry

b) low temperature humid

c) room temperature dry

d) boiling humid

Correct answer: b

67. 1.5g solute 15g water saturated solubility

a) 0.1 g/g

b) 10 g/100g

c) 1 g/10g

d) 15 g/100g

Correct answer: b

68. 16g NaOH 100g water mass percentage solute

a) 13.79%

b) 16%

c) 100%

d) 86.21%

Correct answer: a

69. 35ml methanol 65ml water volume percentage

a) 35%

b) 65%

c) 50%

d) 100%

Correct answer: a

70.Solute particles solution diameter

- a) >1000 nm
- b) 1-1000 nm
- c) <1 nm
- d) 100-1000 nm

Correct answer: c

71.Ice cubes drink heterogeneous because

- a) uniform composition
- b) visible different phases
- c) transparent
- d) stable

Correct answer: b

72.Perfume homogeneous mixture

- a) gas liquid
- b) liquid gas
- c) solid liquid
- d) all gases

Correct answer: a

73.Naphthalene benzene non-aqueous

- a) aqueous
- b) solution
- c) suspension
- d) emulsion

Correct answer: b

74.Sodium iodide highest solubility water

- a) 36 g

b) 95 g

c) 184 g

d) 91 g

Correct answer: c

75. Endothermic dissolution solubility temperature

a) decreases

b) increases

c) constant

d) zero

Correct answer: b

76. Exothermic dissolution solubility temperature

a) increases

b) decreases

c) constant

d) doubles

Correct answer: b

77. Aquatic animals prefer cold water higher

a) carbon dioxide

b) dissolved oxygen

c) nitrogen

d) salts

Correct answer: b

78. Carbonated drinks pressure released

a) gas escapes

b) liquid evaporates

c) solute precipitates

d) solution freezes

Correct answer: a

79. Mass percentage w/w means

a) weight volume

b) weight weight

c) volume volume

d) mole mole

Correct answer: b

80. Commercial syrups expressed

a) w/w

b) v/v

c) molarity

d) molality

Correct answer: b